



Illuminating
ENGINEERING SOCIETY

APPROVED METHOD:
**PROJECTING LONG-TERM LUMINOUS
FLUX MAINTENANCE OF LED
LAMPS AND LUMINAIRES**
AN AMERICAN NATIONAL STANDARD



**APPROVED METHOD:
PROJECTING LONG-TERM LUMINOUS
FLUX MAINTENANCE OF LED
LAMPS AND LUMINAIRES
AN AMERICAN NATIONAL STANDARD**

Publication of this report
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared for IES
By the IES Testing Procedures Committee**



Copyright 2020 by the Illuminating Engineering Society.

Approved by the IES Standards Committee February 13, 2020 as a Transaction of the Illuminating Engineering Society.

Approved April 23, 2020 as an American National Standard.

All rights reserved. No part of this publication may be reproduced in any form, in any electronic retrieval system or otherwise, without prior written permission of the IES.

Published by the Illuminating Engineering Society, 120 Wall Street, New York, New York 10005.

IES Standards are developed through committee consensus and produced by the IES Office in New York. Careful attention is given to style and accuracy. If any errors are noted in this document, they should be forwarded to Brian Liebel, Director Standards, at standards@ies.org or the above address for verification and correction. The IES welcomes and urges feedback and comments.

Printed in the United States of America.

ISBN# 978-0-87995-390-4

DISCLAIMER

IES publications are developed through the consensus standards development process approved by the American National Standards Institute. This process brings together volunteers representing varied viewpoints and interests to achieve consensus on lighting recommendations. While the IES administers the process and establishes policies and procedures to promote fairness in the development of consensus, it makes no guaranty or warranty as to the accuracy or completeness of any information published herein.

The IES disclaims liability for any injury to persons or property or other damages of any nature whatsoever, whether special, indirect, consequential or compensatory, directly or indirectly resulting from the publication, use of, or reliance on this document.

In issuing and making this document available, the IES is not undertaking to render professional or other services for or on behalf of any person or entity. Nor is the IES undertaking to perform any duty owed by any person or entity to someone else. Anyone using this document should rely on his or her own independent judgment or, as appropriate, seek the advice of a competent professional in determining the exercise of reasonable care in any given circumstances.

The IES has no power, nor does it undertake, to police or enforce compliance with the contents of this document. Nor does the IES list, certify, test or inspect products, designs, or installations for compliance with this document. Any certification or statement of compliance with the requirements of this document shall not be attributable to the IES and is solely the responsibility of the certifier or maker of the statement.

AMERICAN NATIONAL STANDARD

Approval of an American National Standard requires verification by ANSI that the requirements for due process, consensus, and other criteria have been met by the standards developer.

Consensus is established when, in the judgment of the ANSI Board of Standards Review, substantial agreement has been reached by directly and materially affected interests. Substantial agreement means much more than a simple majority, but not necessarily unanimity. Consensus requires that all views and objections be considered, and that a concerted effort be made toward their resolution.

The use of American National Standards is completely voluntary; their existence does not in any respect preclude anyone, whether that person has approved the standards or not, from manufacturing, marketing, purchasing, or using products, processes, or procedures not conforming to the standards.

The American National Standards Institute does not develop standards and will in no circumstances give an interpretation to any American National Standard. Moreover, no person shall have the right or authority to issue an interpretation of an American National Standard in the name of the American National Standards Institute. Requests for interpretations should be addressed to the secretariat or sponsor whose name appears on the title page of this standard.

CAUTION NOTICE: This American National Standard may be revised at any time. The procedures of the American National Standards Institute require that action be taken to reaffirm, revise, or withdraw this standard no later than five years from the date of approval. Purchasers of American National Standards may receive current information on all standards by calling or writing the American National Standards Institute.

Prepared by the IES Testing Procedures Committee.

Becky Kuebler, *Chair*

Andrew Jackson, *Vice Chair*

David N. Randolph, *Secretary*

Jianzhong Jiao, *Treasurer*

Members

C. K. Andersen	K. C. Fletcher	P.-C. Hung	P. McCarthy
R. P. Bergin	M. L. Grather	M. Kotrebai	G. McKee
R. S. Bergman	Y. Hiebert	J. E. Leland	C. C. Miller
E. Bretschneider	J. Hospodarsky	S. Longo	E. Radkov
P. Elizondo	J. N. Hulett	J. P. Marella	M. B. Sapcoe

Advisory Members

L. M. Ayers	A. A. Feldman	K. M. Liepmann	T. Schneider
J. Baker	J. Frazer	J. Lockner	A. W. Serres
C. A. Bloomfield	K. J. Hemmi	Y. Ohno	G. A. Steinberg
P.-T. Chou	S. Hua	E. Page	L. Swainston
P. Cruz	G. John	D. Park	J. S. Swiernik
M. Damle	J. Juhasz	E. S. Perkins	A. Thorseth
L. Davis	H. Kashaninejad	M. Piscitelli	R. C. Tuttle
J. J. Demirjian	T. Kawabata	T. W. Rasinski	J. C. Vollers
M. E. Duffy	J. D. Kramer	D. Rogers	J. E. Walker
V. Eberhard	K. C. Lerbs	M. P. Royer	Y. Zong

CONTENTS

1.0	Introduction and Scope	1
1.1	Introduction	1
1.2	Scope	1
2.0	Normative References	1
2.1	ANSI/IES LM-80-20	1
2.2	ANSI/IES TM-21-19	1
2.3	ANSI/IES LM-84-20	1
3.0	Definitions	1
3.1	device under test (DUT)	1
3.2	sample set	1
4.0	Samples and Test Data	1
4.1	Samples	1
4.1.1	Sampling Selection Recommendations	1
4.1.2	Sampling Size Recommendations	2
4.2	Data to be Used	2
4.2.1	Data for Direct Extrapolation	2
4.2.2	Data for Combined Extrapolation	2
5.0	Luminous Flux Maintenance Projection	2
5.1	Procedures for Projection Method 1, Direct Extrapolation of LM-84 Data	2
5.1.1	Normalization	3
5.1.2	Average	3
5.1.3	Data Used for Curve-Fit	3
5.1.4	Curve-Fit	3
5.1.5	Adjustment of Results	3
5.1.6	Notation for Luminous Flux Maintenance Life	4
5.1.7	Ambient Temperature Data Interpolation for Direct Extrapolation	4
5.1.7.1	Select Case Temperature T_s	4
5.1.7.2	Convert All Temperatures to Kelvins	4
5.1.7.3	Use the Arrhenius Equation to Calculate the Interpolated Luminous Flux Maintenance Life	4
5.1.7.4	Applicability of the Arrhenius Equation	5
5.1.7.5	Limit for Extrapolation	5

5.2	Procedures for Projection Method 2, Combined Extrapolation of LM-84 Data and LM-80 Data	5
5.2.1	Normalization	6
5.2.2	Average	6
5.2.3	Selection of the LM-80 Data Set	6
5.2.4	Curve-Fit of the LM-80 Data Set	6
5.2.5	Calculation of the $\Delta\alpha$ Value	6
5.2.6	Projection	6
5.2.7	Adjustment of Results	7
5.2.8	Notation for Luminous Flux Maintenance Life	7
6.0	Report	8
Annex A	Determination of LED Forward Current in the Combined Extrapolation Method	9
Annex B	Derivation of the Formulas Used for Combined Extrapolation Method	9
Annex C	Example of the Combined Extrapolation Method	11
Annex D	Statistical Modeling	13
Informative References		15

1.0 Introduction and Scope

1.1 Introduction

The objective of this Technical Memorandum is to provide guidance and recommended procedures for sampling, test intervals and duration, and a method for long term luminous flux maintenance projection for LED lamps and luminaires. The intent is to help product manufacturers and users, standards developing bodies, and other organizations to avoid any unnecessary burdens related to excessive product testing.

A balance was sought between testing time and effort on one side, and sufficient statistical rigor on the other. A direct and a combined method for luminous flux maintenance projection are proposed; the latter allows the use of luminous flux maintenance data obtained for both the lamp or luminaire being tested, and the LED sources used in it.

Typically, test data for LED sources are available for at least 6,000 hours prior to the design of lamps or luminaires with them. This provides the starting information for the luminous flux maintenance projection according to the combined method. That method also takes into account any detectable further contribution from the remaining lamp or luminaire components, rather than using only LED-source test data as a proxy. As a result, the luminous flux maintenance projection for the lamp or luminaire can begin as early as 3,000 hours into the testing when the protocol proposed by this document is followed.

1.2 Scope

This document recommends the methods for projecting long-term luminous flux maintenance of LED lamps and luminaires using data obtained when testing them per *ANSI/IES LM-84-20, Approved Method: Measuring Optical Radiation Maintenance of LED Lamps, Light Engines, and Luminaires*, as well as data when testing LED sources per *ANSI/IES LM-80-20, Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays, and Modules*.

2.0 Normative References

2.1 ANSI/IES LM-80-20

Illuminating Engineering Society. Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays, and Modules. New York: IES; 2020.

2.2 ANSI/IES TM-21-19

Illuminating Engineering Society. Technical Memorandum: Projecting Long-Term Lumen, Photon, and Radiant Flux Maintenance of LED Light Sources. New York: IES; 2019.

2.3 ANSI/IES LM-84-20

Illuminating Engineering Society. Approved Method: Measuring Optical Radiation Maintenance of LED Lamps, Light Engines, and Luminaires. New York: IES; 2020.

3.0 Definitions

3.1 device under test (DUT)

An LED lamp or luminaire under test.

3.2 sample set

The plurality of LED lamps or luminaires being tested under a given test condition.

4.0 Samples and Test Data

4.1 Samples

4.1.1 Sampling Selection Recommendations.

Commercial users of the information and analysis resulting from this document will, in general, assume that the data are “production representative” unless otherwise stated. “Production representative” means produced using production materials, with production processes and production assembly personnel, though not necessarily in a full production mode (for example, the units may be from a production pilot run). Not all testing is intended to serve the same purpose. This document provides a standardized method that may be appropriate.

Manufacturers may have a need to test in the developmental stages of product design programs, or to test “production representative” versions of a product. Samples for testing shall be selected based on a documented rationale that is consistent with the test intent. Whenever samples are not “production representative,” the ANSI/IES TM-28-20 report shall state the exceptions.

4.1.2 Sampling Size Recommendations. The required minimum number of samples of a specific DUT tested at each of the specified operational and environmental conditions is 3 for the “direct” extrapolation method (see **Section 5.1**) and 5 for the “combined” extrapolation method (see **Section 5.2**). The total length of the projection allowed depends on the number of samples used (please refer to the corresponding Adjustment of Results section in each method).

4.2 Data to be Used

All data from the sample set at a given temperature and forward current for the DUTs shall be collected according to ANSI/IES LM-84-20 (see **Section 2.2**) for luminous flux maintenance life projection (henceforth referred to as “LM-84 data”).

An average forward current value and a case temperature, T_s , value (as defined in ANSI/IES LM-80-20) shall be provided by the test requestor for the LED packages or LED modules inside the lamps or luminaires, whenever the combined extrapolation method is used.

In order to use the procedures listed in **Section 5.1.7.1** and **Section 5.2**, the in-situ LED case temperature, T_s , in the DUT shall be obtained (refer to Annex A of ANSI/IES LM-84-20).

Data collected within a ± 48 -hour window of each “1,000-hour measurement point”—e.g., from 952 hours to 1,048 hours—shall be acceptable as a “1,000-hour reading” and shall be used for luminous flux maintenance projection with the actual measured elapsed time value. The ± 48 -hour window is also applicable to intervals smaller than 1,000 hours. The actual measured elapsed time values shall be used in the projection calculation.

The data collection points used in the projection

calculation shall be equally dispersed in time (to within ± 48 hours). No two consecutive data collection intervals after the initial 1,000 hours shall differ by more than 96 hours in length. For example, data may be used in the projection calculation that are collected every 1,000 hours (± 48 hours), or every 500 hours (± 48 hours), but not every 1,000 hours and occasionally at 500 hours, since this would give excessive statistical weight to certain data points.

4.2.1 Data for Direct Extrapolation. LM-84-20 data shall be collected initially and at least once every 1,000 hours of operating time (to within ± 48 hours) thereafter, for the duration of the test but over no fewer than 6,000 hours of total operating time. Data collected at intervals greater than 1,000 hours shall not be used.

4.2.2 Data for Combined Extrapolation. LM-84-14 data were taken for at least 3,000 hours but less than 6,000 hours. LM-84-20 data shall be collected initially, at 1,000 hours (to within ± 48 hours) of operating time, and at least once every 500 hours (to within ± 48 hours) of operating time thereafter, until 6,000 hours of total operating time or the end of the test, whichever comes first. Luminous flux maintenance data of LED packages or LED modules needed for combined extrapolation shall be collected according to ANSI/IES LM-80-20 (henceforth referred to as “LM-80 data”) and processed according to ANSI/IES TM-21-19.

5.0 Luminous Flux Maintenance Projection

5.1 Procedures for Projection Method 1, Direct Extrapolation of LM-84 Data

This method is recommended for all cases where at least 6,000 hours’ worth of ANSI/IES LM-84-20 data are available. It uses a curve-fit to the collected data to extrapolate the luminous flux maintenance value to the time point where the luminous flux decreases to the minimum acceptable level (for example, 70% of initial luminous flux). That time point is the projected luminous flux maintenance life. The same curve-fit of the collected data can also be used to determine the

projected luminous flux at given future time points (e.g., 25,000 hours or 35,000 hours).

The method is applied separately for each set of DUT test data collected at each operational (e.g., drive current) and environmental (e.g., ambient temperature) condition, as specified in ANSI/IES LM-84-20, using the following procedure. For intermediate environmental conditions, the data can be interpolated as specified in **Section 5.1.7**.

5.1.1 Normalization. All collected data are normalized to a value of 1 (100%) at 0 hours for each device tested.

5.1.2 Average. The normalized measured data of all samples within the same data set defined in **Section 5.1** are averaged for each test condition at each measurement point. The following sub-sections apply to the test condition under consideration.

5.1.3 Data Used for Curve-Fit. For data sets of test duration, D , from 6,000 hours up to 10,000 hours, the data used for the curve-fit shall be the last 5,000 hours of data. Data before the reading at 1,000 hours shall not be used for the curve fit.

For data sets of test duration greater than 10,000 hours, the data for the last 50% of the total test duration shall be used for the curve-fit. In other words, all data points between $D/2$ and D shall be used. For example, if the test duration is 13,000 hours, all data points between 6,500 hours and 13,000 hours are used. If there is no data point at $D/2$, then the next-lower time point shall be included in the data fitting. For example, for D of 13,000 hours of data taken every 1,000 hours, the data points between 6,000 hours and 13,000 hours are used.

5.1.4 Curve-Fit. An exponential least-squares curve-fit through the averaged values is performed, as specified in the preceding section, using **Equation 5-1**:

$$\Phi(t) = Be^{-\alpha t}, \quad (5-1)$$

where:

t = operating time, in hours

$\Phi(t)$ = averaged normalized luminous flux output at time t

B = projected initial constant derived by the

least-squares curve-fit of the ANSI/IES LM-84-20 data

α = decay rate constant derived by the least-squares curve-fit of the ANSI/IES LM-84-20 data

Equations 5-2 and 5-3 are used to project the luminous flux maintenance life:

$$L_{70} = [\ln(B/0.7)]/\alpha \quad (5-2)$$

$$L_{50} = [\ln(B/0.5)]/\alpha \quad (5-3)$$

For any levels of luminous flux maintenance, use the generic form of the equation:

$$L_p = [\ln(100B/p)]/\alpha, \quad (5-4)$$

where:

L_p = luminous flux maintenance life, in hours

p = the percentage of initial lumen output that is maintained

When $\alpha > 0$, then the exponential curve-fit decays to zero, and L_p is positive. When $\alpha < 0$, then the exponential curve-fit increases over time, and L_p is negative.

Whenever an L_p value is reached experimentally in the course of ANSI/IES LM-84-20 testing, the reported value shall be obtained by linear interpolation between the two nearest test points and takes precedence over any value projected by the formulas above.

5.1.5 Adjustment of Results. For any given sample size of at least 3 lamps or luminaires, luminous flux values shall not be projected beyond x times the total test duration (in hours) of measured data, where x is a multiplication factor. The values of x for different sample sizes are listed in **Table 5-1**.

Table 5-1. Values of the Multiplication Factor x for Different Sample Sizes Using the Direct Extrapolation Method

Sample Size	Multiplication Factor, x
3	3
4	4
5 – 6	5
7 – 9	5.5
≥ 10	6

When the calculated luminous flux maintenance life (e.g., L_{70}) is positive and less than or equal to x times the total test duration, the calculated luminous flux maintenance life is the reported luminous flux maintenance life.

When the calculated luminous flux maintenance life (e.g., L_{70}) is positive and greater than x times the total test duration, the reported luminous flux maintenance life value is limited to x times the test duration.

When the calculated luminous flux maintenance life (e.g., L_{70}) is negative, the reported luminous flux maintenance life is x times the total test duration, and any projections of normalized lumen output at specific operating times beyond the duration of the tests shall be reported to be equal to the average normalized lumen output at the last measurement point.

5.1.6 Notation for Luminous Flux Maintenance Life.

The luminous flux maintenance life projected in this method shall be expressed using the notation:

$$L_p(Dk),$$

where p is the percentage of initial lumen output that is maintained, and D is the total duration of the test in hours divided by 1,000 and rounded to the nearest integer. Examples:

$$L_{70}(6k) \text{ for 6,000 hours' test data}$$

$$L_{70}(10k) \text{ for 10,000 hours' test data}$$

If the calculated L_p value is reduced by the x -times rule from **Section 5.1.5**, the luminous flux maintenance life value shall be expressed with a symbol ">". Example:

$$L_{70}(6k) > 36,000 \text{ hours (at } T_a = 25^\circ\text{C)}$$

If the L_p value is reached experimentally in the course of ANSI/IES LM-84-20 testing, then the luminous flux maintenance life shall be expressed with the D value to be equal to the experimental average L_p value in hours divided by 1,000 and rounded to the nearest integer. Example:

$$L_{70}(4k) = 4,400 \text{ hours (at } T_a = 25^\circ\text{C)}$$

5.1.7 Ambient Temperature Data Interpolation for Direct Extrapolation. When the in-situ DUT's ambient temperature, T_i , is different from the temperatures used

for ANSI/IES LM-84-20 tests (e.g., 25°C and 45°C), the following procedures should be used to predict luminous flux maintenance life of the DUTs corresponding to the in-situ case temperature with the same or lower operational condition (e.g., forward current).

5.1.7.1 Select Case Temperature T_s . The in-situ LED case temperatures, $T_{s,i}$, used for in-situ luminous flux maintenance life interpolation shall fall between the closest lower case temperature, $T_{s,1}$, and the closest higher case temperature, $T_{s,2}$, to the in-situ $T_{s,i}$ temperature to be interpolated.

5.1.7.2 Convert All Temperatures to Kelvins. The following formula shall be used to convert the temperatures from degrees Celsius to kelvins (K):

$$T[\text{K}] = T[^\circ\text{C}] + 273.15 \quad (5-5)$$

Only values in kelvins shall be used in the calculations shown in the following sections.

5.1.7.3 Use the Arrhenius Equation to Calculate the Interpolated Luminous Flux Maintenance Life. The Arrhenius equation (Equation 5-6) shall be used to calculate in-situ decay rate constant α_i :

$$\alpha_i = A \exp(-E_a/(k_B T_{s,i})) \quad (5-6)$$

where:

A = pre-exponential factor

E_a = activation energy, in eV

$T_{s,i}$ = in-situ case temperature, in K

k_B = Boltzmann's constant (8.6173×10^{-5} eV/K)

The following intermediate calculation steps need to be carried out in order to find the decay rate constant α_i at in-situ case temperature $T_{s,i}$ between $T_{s,1}$ and $T_{s,2}$.

Step 1: Obtain α_1 and α_2 for the two in-situ case temperatures $T_{s,1}$ and $T_{s,2}$, which were calculated in the curve-fits performed per **Section 5.1.4**. Calculate the ratio of E_a/k_B :

$$\frac{E_a}{k_B} = \frac{\ln \alpha_1 - \ln \alpha_2}{(1/T_{s,2}) - (1/T_{s,1})}, \quad (5-7)$$

Step 2: Using the above ratio, plug in $T_{s,1}$ to calculate pre-exponential factor A :

$$A = \alpha_1 \cdot \exp(E_a/k_B T_{s,1}) \quad (5-8)$$

Step 2 can also be used for substituting α_1 and $T_{s,1}$ with α_2 and $T_{s,2}$

Step 3: Calculate B_0 :

$$B_0 = \sqrt{B_1 B_2}, \quad (5-9)$$

where:

B_1 = the projected initial constant for lower T_s case temperature

B_2 = the projected initial constant for higher T_s case temperature

Step 4: Using the above result for B_0 , calculate luminous flux maintenance life L_p for in-situ case temperature $T_{s,i}$:

$$L_p = \ln(100B_0/p)/\alpha_i \quad (5-10)$$

Results from the above calculation steps as well as the parameters used in the calculations are to be reported, as shown in **Table 1** in the Report section.

Step 5: Calculate the in-situ luminous flux output $\phi(t)$ at time t for $T_{s,i}$:

$$\Phi(t) = B_0 \cdot \exp(-\alpha_i t) \quad (5-11)$$

5.1.7.4 Applicability of the Arrhenius Equation. The Arrhenius equation can be used only if both decay rate constants α_1 and α_2 have positive values. In cases where one or both α values are negative (i.e., increasing luminous flux over time), a conservative projection shall be used as specified in **Section 5.2.7**.

If only one α value is positive, the reported luminous flux maintenance projections and L_p values shall be used for T_i .

If neither α value is positive, the reported luminous flux maintenance life, L_{70} will be x times the total test duration of measured data per **Table 5-1** (see **Section 5.1.5**), and any projections of normalized lumen output at specific operating times beyond the duration of the tests shall be reported to be equal to the lower normalized lumen output at the last measurement point between T_1 and T_2 .

5.1.7.5 Limit for Extrapolation. Extrapolation of any L_p value above the highest ambient operating temperature used in the LM-84-20 test shall not be performed. For example, if the highest LM-84-14 ambient operating temperature is 45 °C, an L_p at 55 °C shall not be extrapolated.

If the actual ambient operating temperature is below the lowest such temperature used in the LM-84-20 test, the reported L_p value shall be using the lowest temperature in the LM-84-20 test. For example, if the lowest LM-84-20 ambient test temperature is 25 °C, an L_p at 15 °C shall use L_p at 25 °C.

For interpolation between case temperatures with test durations of unequal length, in instances where the higher temperature has a shorter testing duration, projections shall not extend beyond x times the test duration at the higher temperature, where x is specified in **Table 5-1** (see **Section 5.1.5**). In instances where the lower temperature has a shorter testing duration, projections shall not extend beyond x times the test duration at the higher temperature, where x is specified in **Table 5-1**.

5.2 Procedures for Projection Method 2, Combined Extrapolation of LM-84 Data and LM-80 Data

This method is recommended only for cases where at least 3,000 hours' but fewer than 6,000 hours' worth of LM-84 data are available for the lamps or luminaires tested, and at least 6,000 hours' worth of LM-80 data are available for the LED packages or modules used in the DUTs, provided the latter data meet the following requirement:

*The LM-80 data used in this method shall contain at least one data set obtained at the same or higher T_s , as compared to the DUT's T_s value provided in **Section 4.2**, and at the same or higher forward current, as compared to the forward current value provided in **Section 4.2**, for the LED packages or modules used in the DUTs.*

The method uses a projection from a selected LM-80 data set, based on ANSI/IES TM-21-19 and corrected for additional luminous flux decay in the lamp or luminaire if observed in the LM-84 data, as described further below.

5.2.1 Normalization. All collected IES LM-84-20 data are normalized to a value of 1 (100%) at 0 hours for each device tested.

5.2.2 Average. The normalized measured LM-84-20 data of all samples within the same data set defined in **Section 4.2.2** are averaged for each test condition at each measurement point. The following sub-sections apply to the test condition under consideration.

5.2.3 Selection of the LM-80 Data Set. From the available LM-80 data, the data set obtained at the same or closest higher T_s , as compared to the DUT in-situ T_s value obtained per Annex A in ANSI/IES LM-84-20, and at the same or closest higher forward current as compared to the forward current value for the LED packages or LED modules in the DUTs provided in **Section 4.2.2**, is selected.

5.2.4 Curve-Fit of the LM-80 Data Set. A least-squares curve-fit is performed on the selected LM-80 data set, as described in ANSI/IES TM-21-19.

5.2.5 Calculation of the $\Delta\alpha$ Value. The following steps are used to calculate $\Delta\alpha$.

Step 1: Take the averaged normalized LM-84-20 luminous flux maintenance data points for the test condition of interest.

Step 2: Discard any data obtained before 1,000 hours.

Step 3: Find the luminous flux maintenance data points for the same test times (to within ± 48 hours) from the LM-80 data set selected as described in **Section 5.2.3**. Whenever the same test times are unavailable in the LM-80 data, obtain values corresponding to the same test times by linear interpolation of the nearest two LM-80 data points.

Step 4: Calculate the ratio for each data point in Step 3 divided by the value in Step 1 corresponding to the same test time (to within ± 48 hours).

Step 5: Calculate the natural logarithm for each ratio obtained in Step 4.

Step 6: Perform a least squares curve-fit using **Equation 5-12**:

$$\ln[r(t)] = \Delta\alpha \cdot t + b, \quad (5-12)$$

where:

$r(t)$ = set of ratios obtained for various times in Step 4

t = the operating time, in hours

$\Delta\alpha$ = the slope

B = the intercept

The formula shown above is derived in **Annex C**. The physical meaning of $\Delta\alpha$ is the difference in exponential lumen decay rate constants observed between the LM-84 data and the LM-80 data.

If $\Delta\alpha = 0$, the LM-84 data under consideration are showing the same luminous flux decay rate (i.e., same luminous flux maintenance) as the selected LM-80 data.

If $\Delta\alpha < 0$, the LM-84-20 data under consideration are showing a slower luminous flux decay rate (i.e., better luminous flux maintenance) than the selected LM-80 (LED, array, or module) data. In this case, $\Delta\alpha$ is set to 0, and b as obtained above is used for the next section. This amounts to assuming the same luminous flux decay rate for the lamp or luminaire data as for the LED, array, or module data. It is to provide a conservative projection based on short-term LM-84-20 data, which may have not yet started to exhibit longer-term luminous flux depreciation to a full extent.

If $\Delta\alpha > 0$, the LM-84 data under consideration are showing a faster luminous flux decay rate (i.e., worse luminous flux maintenance) than the selected LM-80 data. In this case, the latter data need to be corrected for the additional luminous flux depreciation observed in the DUT, as described in **Section 5.2.6**.

5.2.6 Projection. An exponential projection is performed using **Equation 5-13** (see **Annex B** for derivation and **Annex C** for a calculation example).

$$\Phi(t) = \frac{B \cdot \exp[-t(\alpha + \Delta\alpha)]}{\exp(b)}, \quad (5-13)$$

where:

t = operating time, in hours

$\phi(t)$ = projected normalized luminous flux output of the DUT at time t

B = projected initial constant derived by the least-squares curve-fit of the selected LM-80 data per ANSI/IES TM-21-19

α = decay rate constant derived by the least-squares curve-fit of the selected LM-80 data set per ANSI/IES TM-21-19

$\Delta\alpha$ = slope as described in **Section 5.2.5**

b = intercept as described in **Section 5.2.5**

Use **Equations 5-14** and **5-15** to project the luminous flux maintenance life.

$$L_{70} = [\ln(B/0.7) - b] / (\alpha + \Delta\alpha) \quad (5-14)$$

$$L_{50} = [\ln(B/0.5) - b] / (\alpha + \Delta\alpha) \quad (5-15)$$

For any levels of luminous flux maintenance, use the generic form, **Equation 5-16**.

$$L_p = [\ln(100B/p) - b] / (\alpha + \Delta\alpha), \quad (5-16)$$

where:

L_p = luminous flux maintenance life expressed in hours

p is the percentage of initial lumen output that is maintained.

When $(\alpha + \Delta\alpha) > 0$, then the exponential curve-fit decays to zero, and L_p is positive.

When $(\alpha + \Delta\alpha) < 0$, then the exponential curve-fit increases over time, and L_p is negative.

Whenever an L_p value is reached experimentally in the course of LM-84-20 testing, the reported value shall be obtained by linear interpolation between the two nearest test points and takes precedence over any value projected by the formulas above.

5.2.7 Adjustment of Results. For any given sample size of at least 5 DUTs, luminous flux values shall not be projected beyond x times the total test duration of LM-84-20 data available. The values of x for different sample sizes are listed in **Table 5-2**.

When the calculated luminous flux maintenance life (e.g., L_{70}) is positive and less than or equal to x the total test

Table 5-2. Values of the Multiplication Factor x for Different Sample Sizes Using the Combined Extrapolation Method

Sample Size	Multiplication Factor x
5	1.5
6	2
7	2.5
8	3
9	3.5
10	4
11	4.5
12	5
13 – 14	5.5
≥ 15	6

duration, the calculated luminous flux maintenance life value is the reported luminous flux maintenance life value.

When the calculated luminous flux maintenance life (e.g., L_{70}) is positive and greater than x times the total test duration, the reported luminous flux maintenance life value is limited to x times the test duration.

When the calculated luminous flux maintenance life (e.g., L_{70}) is negative, the reported luminous flux maintenance life value is x times the total test duration, and any projections of normalized lumen output at specific operating times beyond the duration of the tests shall be reported to be equal to the average normalized lumen output at the last measurement point.

5.2.8 Notation for Luminous Flux Maintenance Life.

The lumen maintenance life projected in this method shall be expressed using the notation:

$$L_p(D_{\text{lamp}} k, D_{\text{LED}} k),$$

where:

p = the percentage of initial lumen output that is maintained

D_{lamp} = the total duration of the test of the DUT, in hours divided by 1,000 and rounded to the nearest integer

D_{LED} = the total duration of LM-80 data (for LED packages or modules), in hours divided by 1,000 and rounded to the nearest integer

For example, $L_{70}(3k,6k)$ represents DUT test data of 3,000 hours and 6,000 hours LM-80 data.

If the calculated L_p value is reduced by the x -times rule, the luminous flux maintenance life shall be expressed with a symbol ">". Example: $L_{70}(3k,6k) > 18,000$ hours (at $T_a = 25^\circ\text{C}$).

6.0 Report

The report of luminous flux maintenance life projection shall include the following information shown in **Table 6-1**. The calculated and reported L_{70} values shall be rounded to 3 significant digits. All α and B values shall be rounded to 4 significant digits.

When interpolation between sets of test data at different temperatures is used, the additional following information shall be presented, as shown in **Table 6-2** (refer to **Section 5.1.7.1** for case temperature).

Table 6-2. Recommended Information to Be Included in the Report for Interpolation

Metric	Value
$T_{s,1}$ ($^\circ\text{C}$)	
$T_{s,1}$ (K)	
α_1	
B_1	
$T_{s,2}$ ($^\circ\text{C}$)	
$T_{s,2}$ (K)	
α_2	
B_2	
E_a/k_B	
A	
B_0	
$T_{s,i}$ ($^\circ\text{C}$)	
$T_{s,i}$ (K)	
α_i	
Reported $L_{70}(Dk)$	

Table 6-1. Recommended Information to Be Included in the Report at Each Test Condition

Description of LED light source tested (e.g., manufacturer, model, catalog number)	Value
Sample size	
Number of failures	
DUT forward current used in the test	_____ mA
Test duration	_____ hours
Test duration used for projection	_____ hour to _____ hour
Tested case temperature	_____ $^\circ\text{C}$
Rationale if samples are not production-representative (state any exceptions)	

For cases where the mean luminous flux maintenance value of interest is crossed experimentally:	Value
L_p of interest ($p < 100$)	L _____
Interpolated test duration at L_p	_____ hours

For cases where at least 6,000 hours of test data are available:	Value
α test data	
B test data	
Reported $L_{70}(Dk)$	_____ hours

For cases where less than 6,000 hours but at least 3,000 hours of test data are available:	Value
LM-80 data α	
LM-80 data B	
LM-80 data calculated $L_{70}(Dk)$	_____ hours
$\Delta\alpha$ value	
Reported $L_{70}(Dk)$	_____ hours

Annex A – Determination of LED Forward Current in the Combined Extrapolation Method

LED forward current, I_f , is critical in properly assessing performance for the combined extrapolation method. The LED luminous flux, efficacy, chromaticity, lumen depreciation, and other parameters are determined from the actual forward current. The methods in this document rely upon an accurate LED forward current being operated as one of the key comparison elements; inaccurate currents may lead to significantly shorter or longer depreciation projections and limit the confidence of real-world performance.

ANSI/IES LM-80-20 requires precise input current regulation in order to ensure consistent results and to minimize any current variation on the reported measurements. The regulated current is the current at the light source and in most cases can be simply translated into LED I_f . Actual LED I_f in a lamp or luminaire is a result of the system design, including the use of parallel, series, or matrix device topologies, use of supplemental devices (e.g., resistors, diodes), and on-board, integral, or remote power supplies or drivers.

There are numerous techniques for determining the LED forward current to be used in implementing the methods of this document. The most accurate value could be obtained by measuring the current at each LED (or a representative sample) and calculating the average; however, because current has to be measured in series with the LED, this is highly impractical. It is therefore recommended that a combination of measurement and calculation be utilized to derive the best estimated average value.

The first step is to measure the total system current at the output of any control electronics (power supply or driver). For systems with multiple LED packages, LED modules, or LED boards, the next step is to measure the *array or board input current* for each array or board in the lamp or luminaire. For systems with only one LED array or LED board, the *total system current* should be used for the *board input current*. For each array or board, each

LED forward current is calculated through an analysis of the circuiting of the board. Each calculated array or board current is used to calculate the average LED forward current for the system.

A double-check can be completed by performing a total-system analysis by substituting the *average LED forward current* for each LED in the system and calculating the total current required. This value should be roughly equivalent to the measured total system current.

Annex B – Derivation of the Formulas Used for Combined Extrapolation Method

The relationship between the luminous flux maintenance of a lamp (or luminaire) and that of the LED packages (or LED modules) used in it can be expressed as:

$$\Phi(t)_{\text{LM-84}} = \Phi(t)_{\text{LM-80}} \cdot C(t) \quad (\text{B-1})$$

where:

$\phi(t)_{\text{LM-84}}$ = the luminous flux maintenance of a lamp (or luminaire) as measured according to LM-84

$\phi(t)_{\text{LM-80}}$ = the luminous flux maintenance of the LED packages (or modules) used for the lamp (or luminaire) as measured according to LM-80

$C(t)$ = the correction function relating the LM-84 data to the LM-80 data over the operating time

The approach used in the series of calculations beginning with **Equation B-2** determines experimentally the correction function $C(t)$ from available “short term” (less than 6,000 hours) LM-84-20 and LM-80 data, and then uses it on the “long term” (6,000 hours or more) LM-80 data to extrapolate the luminous flux maintenance of the lamp (or luminaire).

According to the approach used in this document and in ANSI/IES TM-21-19, luminous flux maintenance in each case is approximated as a simple exponential decay:

$$\Phi(t)_{\text{LM-84}} = B_{\text{LM-84}} \cdot \exp(-t\alpha_{\text{LM-84}}) \quad (\text{B-2})$$

and

$$\Phi(t)_{\text{LM-80}} = B_{\text{LM-80}} \cdot \exp(-t\alpha_{\text{LM-80}}) \quad (\text{B-3})$$

Solving for $C(t)$ and substituting the latter two formulas in **Equation B-1**:

$$C(t) = \frac{\Phi(t)_{\text{LM-84}}}{\Phi(t)_{\text{LM-80}}} = \frac{B_{\text{LM-84}} \cdot \exp(-t\alpha_{\text{LM-84}})}{B_{\text{LM-80}} \cdot \exp(-t\alpha_{\text{LM-80}})} = \frac{B_{\text{LM-84}}}{B_{\text{LM-80}} \cdot \exp(-\Delta\alpha \cdot t)}, \quad (\text{B-4})$$

where:

$$\Delta\alpha = \alpha_{\text{LM-84}} - \alpha_{\text{LM-80}} \quad (\text{B-5})$$

After taking the natural logarithm and changing the sign on both sides, the formula for $C(t)$ yields:

$$\ln[r(t)] = \Delta\alpha \cdot t + b, \quad (\text{B-6})$$

where:

t = the operating time in hours

and:

$$r(t) = \frac{\Phi(t)_{\text{LM-80}}}{\Phi(t)_{\text{LM-84}}}$$

and:

$$b = \ln\left(\frac{B_{\text{LM-80}}}{B_{\text{LM-84}}}\right)$$

This formula is used in **Section 5.2.5** to determine $C(t)$ experimentally by a least-squares linear fit of "short term" LM-80 and LM-84-20 data.

Then the "short term" LM-84-20 data can be extrapolated by using $C(t)$ on the "long term" LM-80 data, following **Equation B-4**:

$$\Phi(t)_{\text{LM-84}} = \Phi(t)_{\text{LM-80}} \cdot \exp(-\Delta\alpha \cdot t - b) \quad (\text{B-7})$$

Or (after substituting the ANSI/IES TM-21-19 approximation for $\phi(t)_{\text{LM-80}}$:

$$\Phi(t) = \frac{B \exp[-t(\alpha + \Delta\alpha)]}{\exp(b)}, \quad (\text{B-8})$$

where:

$\Phi(t)$ = projection for the lamp (or luminaire) data

α = decay rate constant derived by the least-squares curve-fit of the LM-80 data per ANSI/IES TM-21-19

B = projected initial constant by the least-squares curve-fit of the LM-80 data per ANSI/IES TM-21-19

By taking the natural logarithm on both sides and rearranging the above formula, one can solve for L_{70} , L_{50} , or L_p :

$$L_{70} = \frac{\ln(B/0.7) - b}{\alpha + \Delta\alpha} \quad (\text{B-9})$$

$$L_{50} = \frac{\ln(B/0.5) - b}{\alpha + \Delta\alpha} \quad (\text{B-10})$$

$$L_p = \frac{\ln(100B_0/p) - b}{\alpha + \Delta\alpha} \quad (\text{B-11})$$

Annex C – Example of the Combined Extrapolation Method

Table C-1. 4,500 Hours' LM-84 Test Data at T_s Temperature of 36.5 °C, Nominal Forward Current of 154 mA

Sample #	Hours									
	0	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500
1	1.000	1.018	1.045	1.038	1.063	1.054	1.056	1.047	1.044	1.038
2	1.000	1.026	1.035	1.042	1.061	1.052	1.050	1.059	1.053	1.050
3	1.000	1.026	1.034	1.042	1.060	1.043	1.042	1.050	1.044	1.033
4	1.000	1.026	1.044	1.051	1.070	1.053	1.054	1.057	1.057	1.051
5	1.000	1.018	1.036	1.046	1.071	1.054	1.063	1.046	1.046	1.037
6	1.000	1.027	1.045	1.051	1.071	1.063	1.068	1.051	1.051	1.043
7	1.000	1.028	1.046	1.044	1.073	1.055	1.053	1.044	1.050	1.035
8	1.000	1.033	1.074	1.048	1.058	1.050	1.054	1.046	1.046	1.030
9	1.000	1.027	1.044	1.043	1.062	1.053	1.066	1.060	1.063	1.060
10	1.000	1.025	1.042	1.042	1.059	1.051	1.058	1.031	1.025	1.017
Average	1.0000	1.0254	1.0445	1.0447	1.0648	1.0528	1.0564	1.0491	1.0479	1.0394

Table C-2. 6,000 Hours' LM-80 Test Data at T_s Temperature of 45 °C, Nominal Forward Current of 350 mA

Sample #	Hours						
	0	1,000	2,000	3,000	4,000	5,000	6,000
1	1.000	0.972	1.014	1.013	0.992	0.978	0.967
2	1.000	1.006	0.999	0.992	0.976	0.974	0.971
3	1.000	0.991	1.004	1.005	0.990	0.978	0.966
4	1.000	1.015	1.012	1.006	0.989	0.987	0.983
5	1.000	1.009	1.003	0.996	0.965	0.963	0.958
6	1.000	1.011	1.008	1.002	0.985	0.978	0.977
7	1.000	1.007	0.995	0.988	0.973	0.976	0.972
8	1.000	1.001	1.008	0.998	0.961	0.977	0.970
9	1.000	1.006	0.998	0.991	0.978	0.979	0.978
10	1.000	1.008	1.023	1.019	1.006	0.992	0.977
Average	1.0000	1.0026	1.0064	1.0010	0.9815	0.9782	0.9719
ln (Average)	0	0.002597	0.006380	0.001000	-0.018673	-0.022041	-0.028502

Table C-3. Combined Method Setup for 4,500 Hours' LM-84 Test Data at T_s Temperature of 36.5 °C, and 6,000 Hours' LM-80 Test Data at T_s Temperature of 45 °C

Time, hours	LM-84	LM-80	$r(t)$	$\ln [r(t)]$
1,000	1.0445	1.0026	0.9599	-0.04094
1,500	1.0447	1.0045	0.9615	-0.03924
2,000	1.0648	1.0064	0.9452	-0.05641
2,500	1.0528	1.0037	0.9534	-0.04776
3,000	1.0564	1.0010	0.9476	-0.05387
3,500	1.0491	0.9913	0.9449	-0.05672
4,000	1.0479	0.9815	0.9366	-0.06546
4,500	1.0394	0.9799	0.9427	-0.05900

Table note: The LM-80 points at 1,500, 2,500, 3,500, and 4,500 hours are obtained by linear interpolation, per **Section 5.2.5**. The input values for the averages and the interpolated LM-80 averages are rounded off to the fourth decimal place.

Table C-4. Least-Squares Curve-Fit Summary of 4,500 Hours' LM-84 Data at T_s Temperature of 36.5 °C, and 6,000 Hours' LM-80 data at T_s Temperature of 45 °C, Combined Method

Metric	Value
$\Delta\alpha$ (slope)	-6.299E-06
b (intercept)	-0.03510
α (per TM-21)	7.441E-06
B (per TM-21)	1.0163
$\alpha + \Delta\alpha^{[1]}$	7.441E-06
$1/\exp(b)$	1.0357
Reported $L_{70}^{[2]}$	>18,000

Table notes:

[1] $\Delta\alpha$ set to 0 since $\Delta\alpha < 0$ (see **Section 5.2.5, Step 6**).

[2] 4 x the LM-84 test duration, rounded off to 3 significant digits (see **Section 5.2.7, Table 5-2**).

Table C-5. Lumen Depreciation Projection, Combined Method

Time, hours	Projection	Time, hours	Projection
5,000	1.014	12,500	0.959
5,500	1.010	13,000	0.956
6,000	1.007	13,500	0.952
6,500	1.003	14,000	0.948
7,000	0.999	14,500	0.945
7,500	0.995	14,500	0.945
8,000	0.992	15,000	0.941
8,500	0.988	15,500	0.938
9,500	0.981	16,000	0.934
10,000	0.977	16,500	0.931
10,500	0.974	17,000	0.928
11,000	0.970	17,500	0.924
11,500	0.966	18,000	0.921
12,000	0.963		

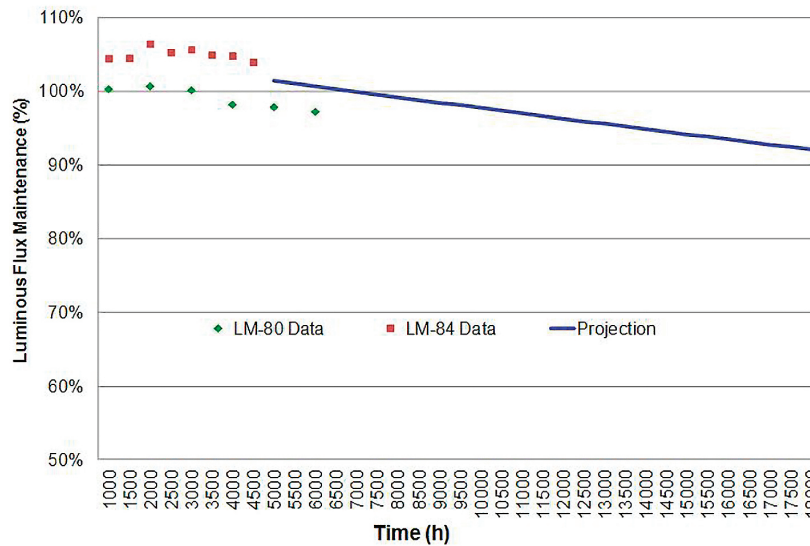


Figure C-1. Plot of the lumen depreciation projection, also showing the averaged experimental LM-84 and LM-80 data used (see Tables C-1, C-2, and C-5).

Annex D – Statistical Modeling

Three different SSL products were used to determine the effect of sample size on lumen degradation estimation. The three products, denoted as "A", "B", and "C", had sample sizes of 20, 10, and 9 LED lamps, respectively. Products A and B were measured for lumen degradation at least every 500 hours, whereas product C was only measured every 1,000 hours. Calculations were performed using only the measured lumen output in the following ways:

- 1,000 to 3,000 hours with an interval of 1,000 hours
- 1,000 to 3,000 hours with an interval of 500 hours
- 1,000 to 6,000 hours with an interval of 1,000 hours
- 1,000 to 6,000 hours with an interval of 500 hours

The data were fitted with a simple exponential model and the lower bound of a 95% prediction interval of the fits was estimated and recorded. For example, 10 lamps measured every 1,000 hours from 1,000 to 3,000 hours were randomly chosen from the 20 lamps of product A. An exponential model was fit to the 10 lamps, and the lower bound estimate of the 95% prediction interval was recorded. The sampling, model fitting, and calculation of the lower bound 95% prediction interval was repeated 100 times. Then this process was repeated for each combination of product (A, B, C) set of applicable sample sizes (20, 19, 15, 10, 9, 8, 7, 6, 5, 4, 3), and applicable measurement lengths and intervals in hours (1,000 to

3,000 every 1,000, 1,000 to 3,000 every 500, 1,000 to 6,000 every 1,000, 1,000 to 6,000 every 500 hours). For each of the mentioned combinations, 100 lower bound 95% prediction interval estimates were calculated and the 10th percentile of these 100 estimates (or the 10th lowest lower bound) was plotted and tabulated, to help determine appropriate extrapolation limit (multiplier) rules for lumen degradation projection. These 10th percentile values may be considered a representative relative worst-case scenario of lumen degradation.

Figure D-1 shows the 10th percentiles of the lower bound of the 95% prediction intervals of the simple exponential model fit to the data. The following conclusions can be drawn:

The general tightness (lack of expected variability) of the 10th percentiles of the lower bound of the 95% prediction interval (excepting products A and B at 1,000 to 3,000 hours) indicates that overall a reduction in sample size does not dramatically reduce the variability between lamps within the same product. Clearly, there is a distinction and hence the need for a methodology here, but there could have been a much more dramatic difference due to sample size. This indicates that the variability between lamps of the same product is sufficiently large that smaller sample sizes can be used and will represent much of that within product variability. This observation was used in the recommendation for the direct projection method, shown below.

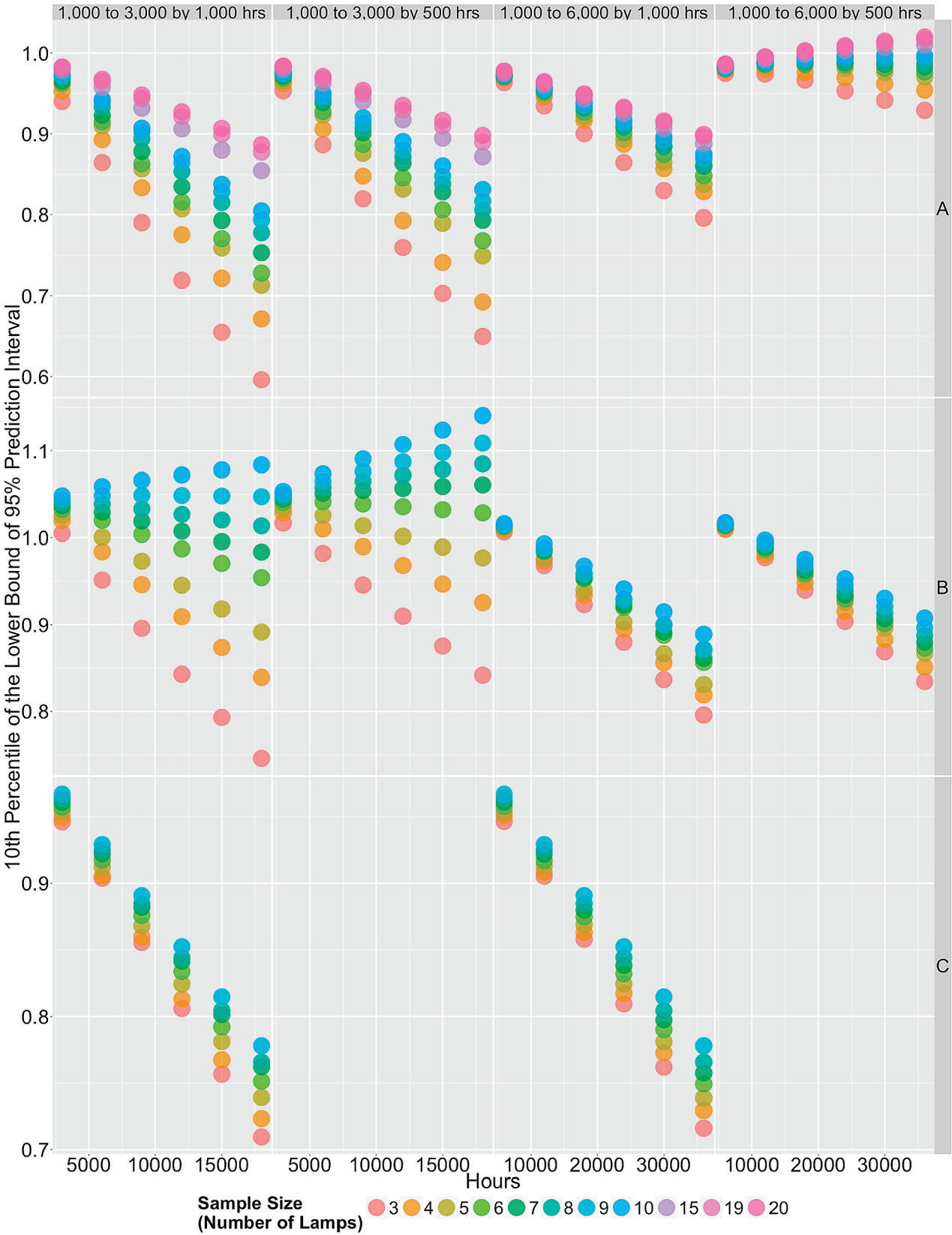


Figure D-1. Results of statistical modeling at different sample sizes.

Another useful conclusion is that there appears to be a larger effect of the total test duration (3,000 hours vs. 6,000 hours), compared to the effect of difference in interval length (500 hours vs. 1,000 hours). For products A and B, the spread of the 10th percentile lower bound prediction interval estimates increased significantly when using only lamps measured from 1,000 to 3,000 hours, compared to when using lamps measured 1,000 to 6,000 hours (regardless of the interval length). The spread or difference of the 10th percentile lower bound prediction interval estimates between sample sizes appears to be only slightly influenced by the interval length, whether 500 or 1,000 hours. In practical terms, this means that different sample sizes would be advisable for data obtained over 3,000 hours (combined method) versus data obtained over 6,000 hours in the direct method of luminous flux projection—significantly higher ones in the former case. This observation was used in the different recommendation for the combined projection method, shown below.

Based on the above conclusions and on the close examination of the differences between the projections at reduced sample sizes compared to those with the largest sample size used in each case, the following recommendations are provided for the maximum length of the projection interval expressed as a multiplier, x , for the LM-84-20 testing duration as a function of sample size for each of the two projection methods used in this document, as shown in **Tables 5-1** and **5-2**. They are intended to maintain the projected lower bounds within 5% of those obtained with the maximum sample size. As in TM-21-19, the maximum projection interval possible in each case is limited by the measurement system uncertainty, in addition to the sample size.

INFORMATIVE REFERENCES

Illuminating Engineering Society. ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: <https://www.ies.org/standards/definitions/>. (Accessed 2019 Aug 29).

Illuminating Engineering Society. ANSI/IES LM-79-19, Approved Method: Optical and Electrical Measurements of Solid-State Lighting Products. New York: IES; 2019.

Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
Affiliation: _____
Address: _____
City: _____ State: _____ Zip: _____ Country: _____
Telephone: _____
Fax: _____
E-mail: _____

I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: _____ Date: _____

2. Title of publications and year published _____

3. Clause (section), sub-clause or paragraph number; and page number: _____

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



Lighting Science Standards

Fundamentals, Metrics and Calculations

Lighting Practice Standards

Design, Engineering, and Specifications

Lighting Applications Standards

Design Criteria and Illumination Recommendations

Lighting Measurements and Testing Procedure Standards

Industry Standardization

Roadway and Parking Facility Lighting Standards

Criteria and Illumination Recommendations

Order# ANSI/IES TM-28-20

ISBN# 978-0-87995-390-4

www.ies.org